

Phase 1 Evaluation

The AI Bidding War: Autonomous Procurement & Vendor Negotiation

Student: Asli Gulcur

Score: **90 / 100**

Score Summary

Dimension	Score	Max
Problem Framing & User Clarity	28	30
Agentic Justification & Initial Architecture	26	30
Scope, Success Criteria & Risk	23	25
Submission Quality & Team Plan	13	15
Total	90	100

Dimension Notes

Problem Framing & User Clarity — 28/30.

The problem is specific and grounded. You name real operational stakeholders, CPOs, supply chain executives, and identify a precise gap. The system will address the challenge that existing procurement SaaS can collect bids but cannot dynamically haggle or weigh multi-dimensional trade-offs in real time. That distinction between static bid collection and active negotiation is a good framing. The only thing missing is a quick quantification of the before-state, even a rough reference to weeks of manual back-and-forth from an industry source or reference would make the ROI case more concrete and easier to defend in Phase 3.

Agentic Justification & Initial Architecture — 26/30.

Your core argument — that a single LLM cannot genuinely negotiate with itself because it collapses context into a simple compromise — is technically correct and specific to this use case. The hub-and-spoke pattern is well chosen and clearly connected to the information isolation requirement. The gap is in the canvas: there is no field capturing how round-by-round state accumulates across the hub, and the Strategic Advisor's role is described at the flow level but not at the architecture level. Its inputs, outputs, and failure behavior deserve their own canvas entry.

Scope, Success Criteria & Risk — 23/25.

The in-scope/out-of-scope section is clean, and the six-scenario evaluation table is a strong piece of work for a Phase 1 submission. The three risks are well-handled — particularly the choice of a deterministic Python guardrail rather than a second LLM check, which is the right governance instinct. The weak spot is criteria precision for the ambiguous scenarios. Scenarios 3, 4, and 5 state “outcome recorded per run” as expected behavior, which is an observation, not

a criterion. Before Phase 2, define what a passing run looks like for each scenario, what would cause you to mark scenario 3 as a governance success versus a negotiation failure?

Submission Quality & Team Plan — 13/15.

The weekly plan is detailed and artifact-specific, you name what gets built and tested each week, which is more useful than a generic task list. The AI usage disclosure is complete and structured with clear distinctions between tools and student intervention, which is exactly what the course policy requires. The one gap is effort estimates: the plan names deliverables but not time allocation per workstream, which makes it harder to identify where schedule risk sits if the prototype phase runs long.

What Worked

The agentic justification is grounded in a technical argument, context contamination, that connects directly to your architectural choice. Most justifications at this phase stay at the level of "agents are more flexible"; yours identifies the specific failure mode of a single-context LLM simulation and explains why separation of concerns solves it. That level of reasoning will carry well into Phase 2's architecture documentation.

The governance design is also a notable strength. Identifying the deterministic Python guardrail as a circuit breaker outside the LLM's probabilistic path is exactly the kind of design decision this course asks for. The layered structure, deterministic guardrail, strategic advisor, human gatekeeper, maps directly to the governance requirements you will need to justify in Phase 3.

3 Things to Address in Phase 2

1. **Define pass/fail criteria for the ambiguous scenarios.** "Scenario 4 states 'outcome recorded per run,' which is an observation, not a criterion. Scenario 3 has a similar gap — 'depending on negotiation' defers the question rather than answering it...". For your Phase 2 evaluation plan, each of the 6 scenarios needs a stated expected outcome plus a measurable condition for pass/fail. For example, Scenario 3 passes if the final awarded price falls between floor and \$12k, or fails and logs NO_AWARD with a documented reason. Without this, your evaluation results in Phase 3 will lack a reference baseline to compare against.
2. **Add the Strategic Advisor to your architecture canvas as a first-class component.** Right now the Advisor appears in the flow description but not in the canvas field structure. Its inputs (scenario metadata, RFP text, serialized thread), outputs (Markdown audit), and failure mode (error stub fallback) need to be documented at the same level of detail as the Vendor and Buyer agents. Phase 2 requires a complete role definition for every agent, and the Advisor is doing substantive governance work that needs to be traceable.
3. **Specify state management across negotiation rounds in your architecture diagram.** The hub-and-spoke pattern isolates vendor threads correctly, but your Phase 2 architecture diagram will need to show how the Buyer's consolidated view is constructed each round — what gets written to state, what gets passed forward, and where round history is stored. If the hub accumulates bid history incorrectly, information isolation breaks down.

References Worth Reading Before Phase 2

1. Kirshner [Wiley Online Library](#), Samuel, Yiwen Pan, Xianghua (Jason) Wu, and Alex Gould. "Talking Terms: Agent Information in LLM Supply Chain Bargaining." *Decision Sciences*, 2025. <https://onlinelibrary.wiley.com/doi/10.1111/deci.70010>
2. Liu, Xianyang, Shangding Gu, and Dawn Song. "AgenticPay: A Multi-Agent LLM Negotiation System for Buyer–Seller Transactions." *arXiv*, 5 Feb. 2026, arXiv:2602.06008. [arXiv https://arxiv.org/abs/2602.06008](https://arxiv.org/abs/2602.06008)
3. Jannelli, V., et al. "Agentic LLMs in the Supply Chain: Towards Autonomous Multi-Agent Consensus-Seeking." *International Journal of Production Research*, 21 Dec. 2025. [Taylor & Francis Online https://www.tandfonline.com/doi/full/10.1080/00207543.2025.2604311](https://www.tandfonline.com/doi/full/10.1080/00207543.2025.2604311)